Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing (01CT1513)		
Lab exp 6	**Date: 23/10/2026**	**Enrolment No:92400133079**

CODE

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.signal import firwin, freqz

# Common parameters
N = 31
window = 'hann'

# Filter specifications
filters = [
    ('FIR Low-Pass Filter', 10000, 3000, 'lowpass'),
    ('FIR High-Pass Filter', 10000, 3000, 'highpass'),
    ('FIR Band-Pass Filter', 20000, [3000, 6000], 'bandpass'),
    ('FIR Band-Stop Filter', 20000, [3000, 6000], 'bandstop')
]

# Design and display each filter separately
for name, Fs, Fc, filter_type in filters:

    # Design FIR filter
    b = firwin(
        numtaps=N,
        cutoff=Fc,
        window=window,
        pass_zero=filter_type,
        fs=Fs
    )

    # Calculate frequency response
    f, H = freqz(b, worN=2048, fs=Fs)

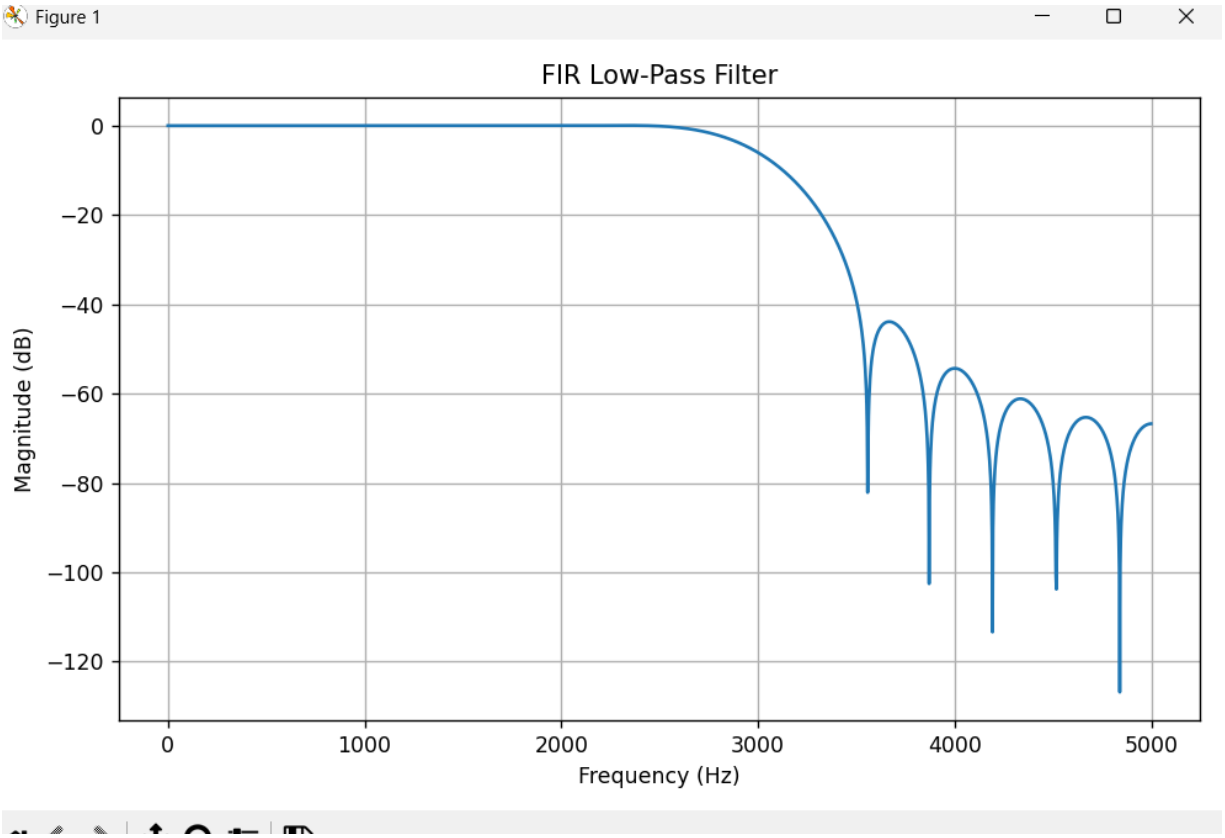
    # Calculate magnitude in dB
    magnitude_db = 20 * np.log10(
        np.maximum(np.abs(H), 1e-10)
    )
```

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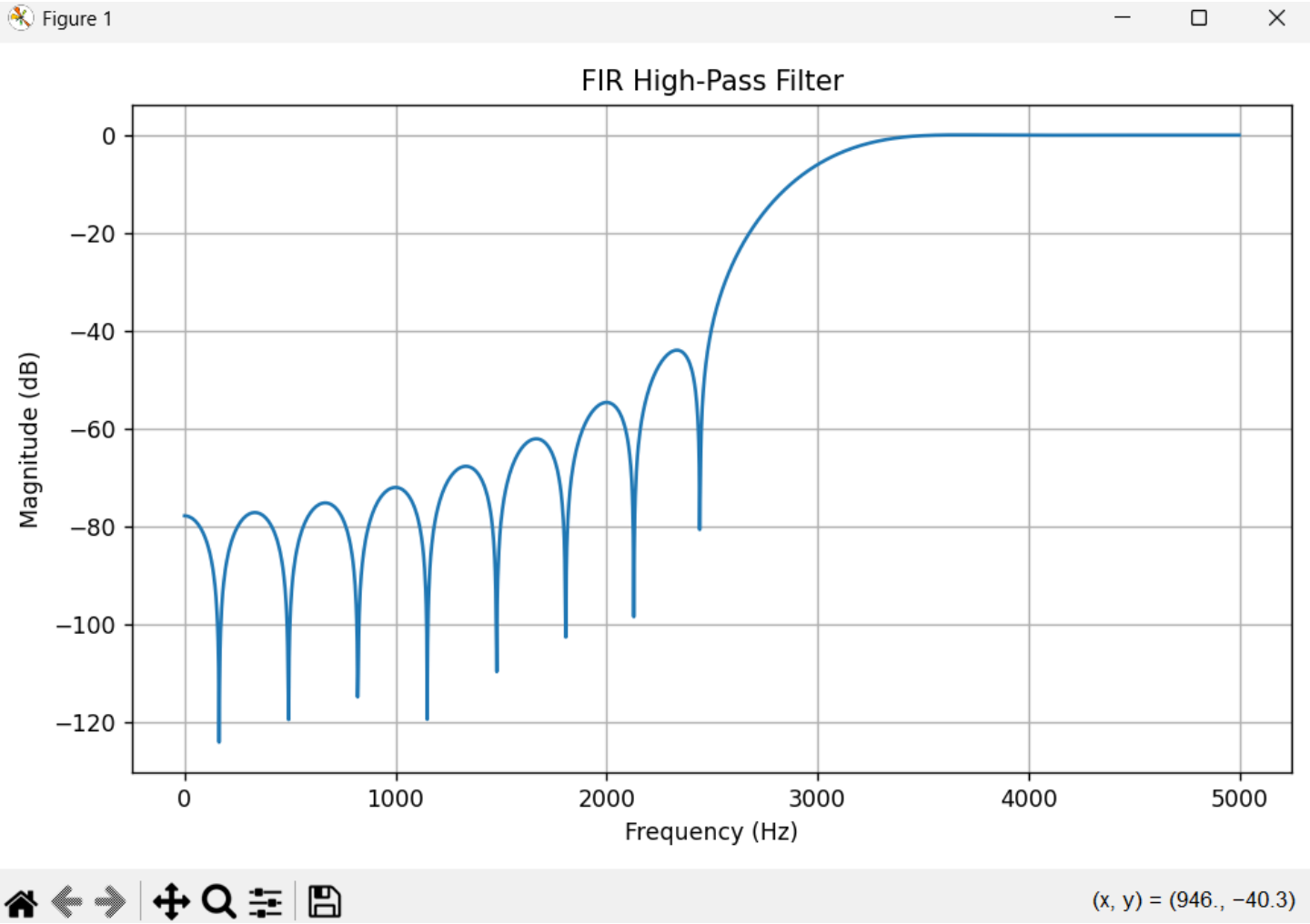
```
# Print filter coefficients
    print("\n" + "=" * 40)
    print(name)
    print("Filter Coefficients:")
    print(b)

    # Plot one filter at a time
    plt.figure(figsize=(8, 5))
    plt.plot(f, magnitude_db)
    plt.title(name)
    plt.xlabel("Frequency (Hz)")
    plt.ylabel("Magnitude (dB)")
    plt.grid(True)
    plt.tight_layout()
    plt.show()
```

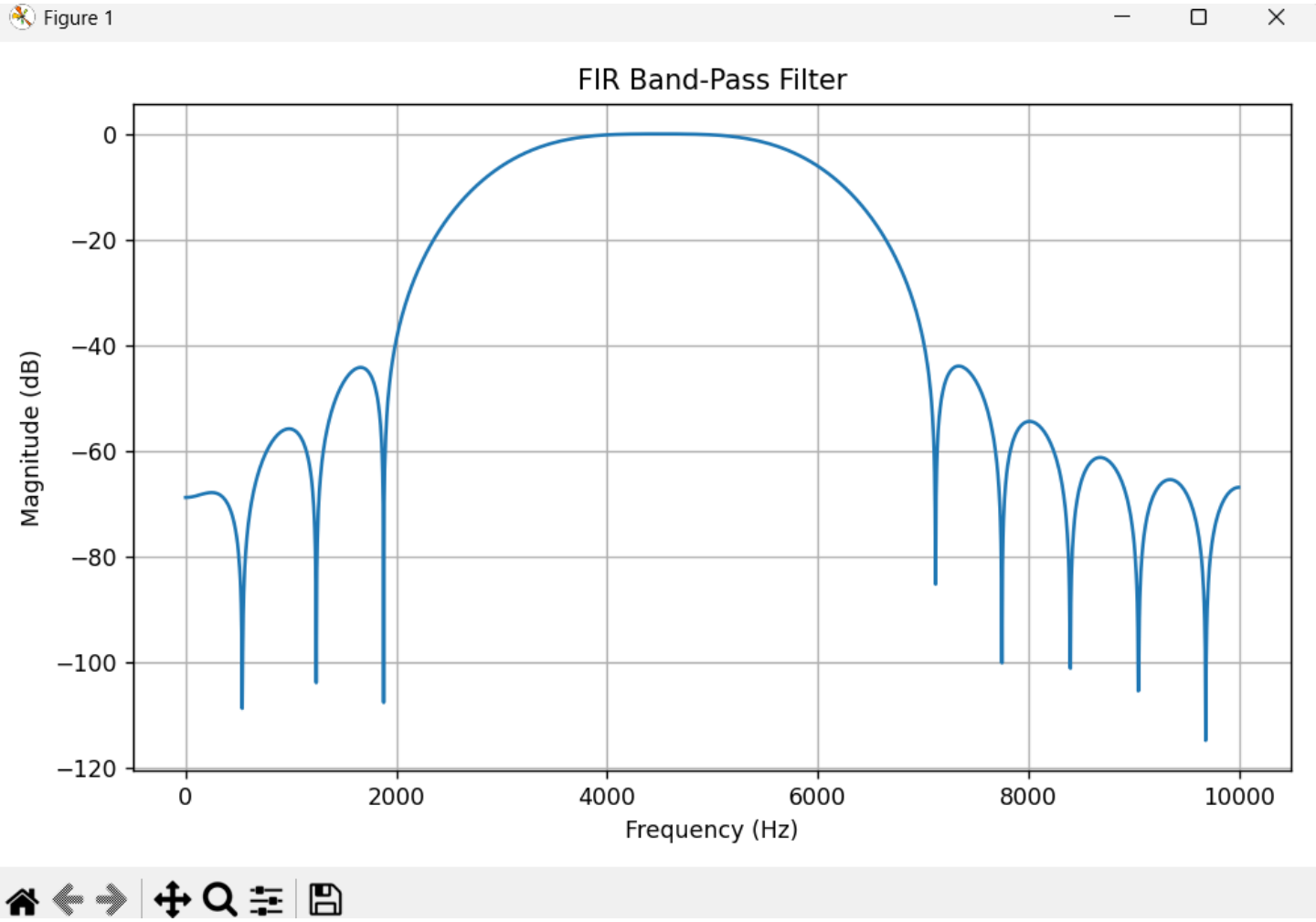
OUTPUT



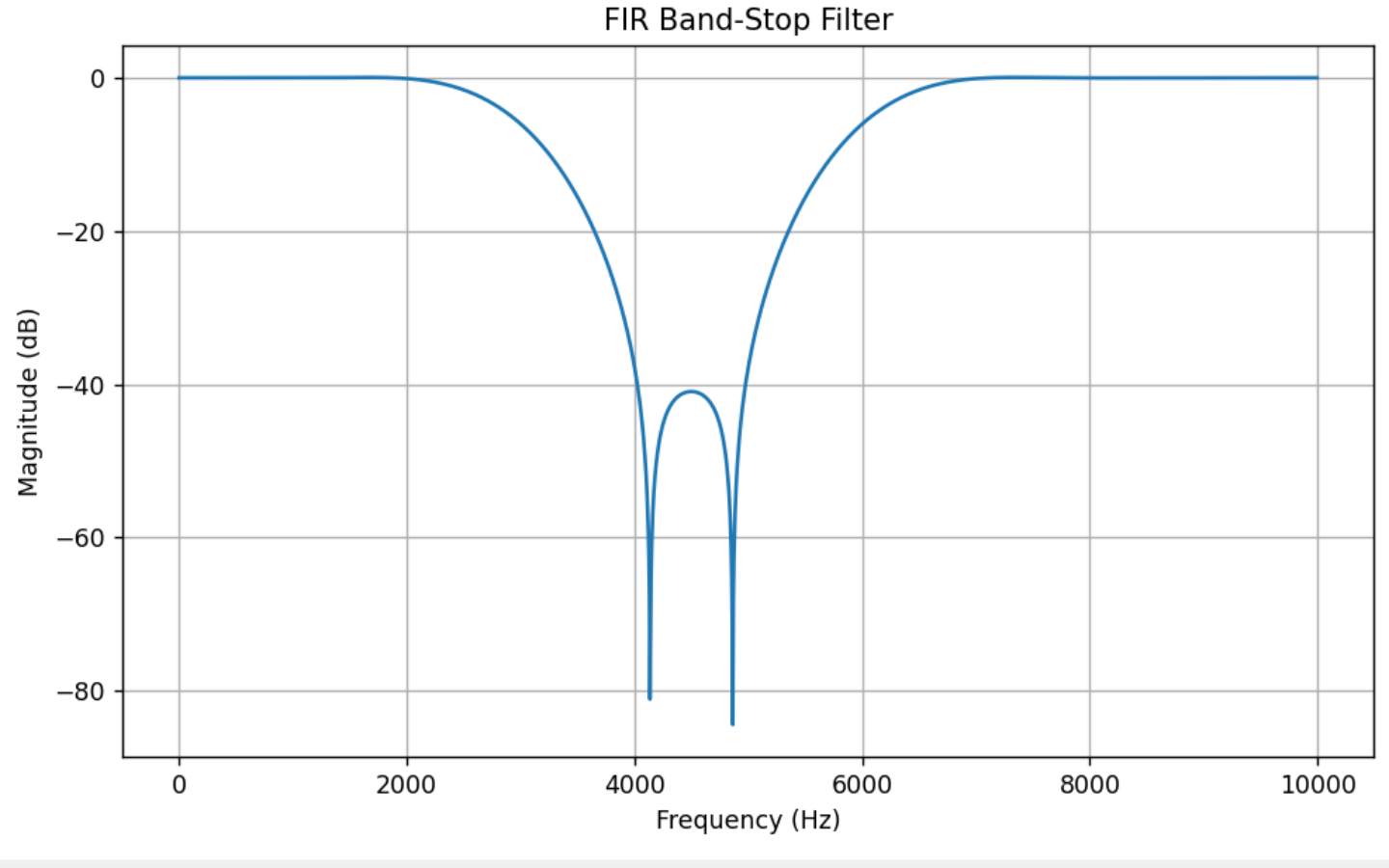
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GITHUB : https://github.com/yugmakwana99/DSIP.git

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